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Kinematic and electromyographic analysis of the Askling L-Protocol for hamstring training

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Abstract

Hamstring injuries are common in field sport athletes. Eccentric strength training of the hamstring muscles is an integral component of rehabilitation programs. The Askling L-Protocol comprises three exercises [extender; diver; glider] that load the hamstrings during eccentric activity. When compared to a conventional exercise-based rehabilitation program, the Askling L-Protocol has been shown to reduce the time to return to sport following hamstring injury and prevalence of injury recurrence. Nevertheless, the mechanisms behind its efficacy have yet to be studied. In this work, we conducted a kinematic and electromyographic analysis of the exercises of the Askling L-Protocol. Eleven healthy individuals performed each of the exercises while electromyographic data from four muscles (including two hamstring muscles) were recorded. Hip and knee angular displacements and velocities were also synchronously recorded using a motion capture system. We found that the L-Protocol elicits a maximal contraction (up to 60% of the MVC in the glider exercise) in the hamstring muscles at a work point similar to the swing phase of running (around 62 degrees hip flexion and 23 degrees knee flexion). No difference in the levels of activation of the hamstrings was observed between the diver and glider, regardless of the different functional role they had in the two exercises. During the extender, the hamstring muscles are stretched and minimally engaged. Finally, co-activation analysis highlighted that through the combination of passive stretch and active eccentric contraction, the hamstrings are trained to co-activate using similar structural modules employed differentially to drive the movement or stabilize it.

KEYWORDS

electromyography, hamstring muscles, kinematics, muscle injuries, sports medicine

1 | INTRODUCTION

Hamstring muscle injuries are one of the most prevalent lower limb musculoskeletal injuries incurred by sprinters and athletes partaking in field sports that are characterized by high-velocity running, accelerations, and decelerations.[1] High incidence rates of hamstring muscle injuries have been reported in Australian rules football,[2-5] association football,[6-8] rugby union,[9] and Gaelic football.[10]

Considering the high incidence rates and prevalence of these injuries in professional- and elite-level field-based sports, prevention and rehabilitation of hamstring muscle injuries are of primary concern among sports medicine physicians and sports physiotherapists. Moreover, hamstring muscle injuries have a reported high recurrence rate,[11] and a strong association has been observed between previous hamstring muscle injury and risk of subsequent injury to this muscle group.[12]

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At the biomechanical level, it has been observed that muscle injuries typically occur when muscles are lengthened to a point beyond the length at which they produce their peak torque.[13] It is established that hamstring muscle injuries are most commonly sustained during high-speed running,[5,14] with the most common site of injury being the long head of the biceps femoris (BF) muscle.[14] Even though the hamstring muscles are substantially loaded during both the stance and swing phases of running gait, it is during the swing phase that they are required to contract forcefully while lengthening at high velocity to their maximal length.[15]

As hamstring injuries are likely to occur as a result of either accumulated or instantaneous eccentric overload, it has been posited that eccentric strength training of hamstring muscles could reduce the overall risk of sustaining a hamstring muscle injury. Eccentric training could in fact be used to increase the number of sarcomeres in series, thus allowing the hamstring muscles to work at longer lengths,[16] thus reducing the amount of work done on the descending and more unstable section of the length-tension curve.[17,18] Such an effect has applications in both prevention and rehabilitation of hamstring muscle injuries.

Eccentric training has been consistently reported to increase the optimum length of tension development[17,19-23] and to be linked with a reduction in hamstring muscle injury risk,[24] as well as an improvement in lower limb strength and sprint performance.[25,26] One of the most commonly utilized exercises for eccentric training of the hamstring muscles is the Nordic hamstring exercise (NHE),[17,27] whose efficacy in reducing hamstring muscle injuries has been substantiated.[6,9,28-30]

Recently, Askling et al[31,32] have proposed an alternative eccentric training protocol for hamstring muscle injuries (originally called and herein referred to as the L-protocol), consisting of exercises focusing on loading the hamstring muscles during extensive lengthening. The L-protocol consists of three eccentric exercises; one aimed at increasing flexibility of the posterior thigh muscles (the extender, EX), one that builds strength by challenging pelvis and trunk stabilization (the diver, DV), and one that loads the hamstrings through controlled lengthening (the glider, GL). Askling and colleagues demonstrated the efficacy of this protocol in two populations of elite athletes, whereby individuals who completed the L-protocol were characterized by a shorter time to return to sport when compared to athletes who completed a similarly scaled conventional training protocol.[31,32] More importantly, Askling and colleagues reported that during a 12-month follow-up, there were no re-injuries among those individuals who participated in the L-protocol. This differs substantially to the typical hamstring muscle re-injury rates reported in the published literature.[33] The efficacy of the L-protocol is primarily believed to lie in the fact that it exercises the hamstring muscles during maximal dynamic lengthening involving movements across both the hip and knee joints. As such, the L-protocol has gained popularity among practitioners. However, to the best of our knowledge, the mechanisms behind its efficacy have yet to be fully elucidated. Accordingly, the aim of this study was to conduct a kinematic and electromyographic analysis of the L-protocol to determine characteristics of muscle activation and co-activation, synchronized with measures of joint displacement and velocity.

2 | METHODS

2.1 | Participants

Eleven individuals (5 females and 6 males, age = 22.2 ± 1.38 years; height = 1.76 ± 0.08 m; body mass = 69.45 ± 8.47 kg) participated in the study. All participants were physically active and regularly engaged in a sports activity (athletics, tag rugby, or hurling), five of them at competitive level, and six of them at recreational level. All of the participants were free from hamstring injuries within the previous 6 months prior to testing. All participants were informed about the background of the study and accepted to participate on a voluntary basis. All data collections took place in the Movement Analysis Laboratory of the School of Public Health, Physiotherapy and Sports Science at UCD. The UCD Human Research Ethics Committee approved the study, and all study procedures were designed and administered accordingly with the Declaration of Helsinki.

2.2 | Experimental setup

During all exercises, surface electromyographic signals (EMG) were recorded from the following muscles: long head of the biceps femoris (BF); semitendinosus (ST); rectus femoris (RF); and gluteus maximus (GM). EMG data were acquired at a sample rate of 2.048 kHz (Biopac Systems, Inc, Goleta, CA, USA). For all participants, EMG data were recorded from the muscles of the right leg. The skin was shaved and cleansed with alcohol before Ag/AgCl bipolar electrodes (Blue Sensor N-00-S, Ambu Medicotest A/S, Ølstykke, Denmark) were placed according to the SENIAM guidelines[34] at an interelectrode distance of 20 mm. Following the placement of the electrodes, which took place at the beginning of each testing session, the participants were required to perform a series of tests to record the EMG during maximal voluntary contractions (MVC) of the target muscles. Specifically, all participants performed three repetitions of knee extension, knee flexion, and hip extension MVC tests using a leg extension machine. Before performing the actual tests, participants were familiarized with the three MVC exercises at low force levels. During all the exercises of the

L-protocol, the movement of the subjects was recorded using a stereophotogrammetric system (Codamotion, Leicestershire, UK). Movement was recorded in the sagittal plane using an ad hoc montage of the active markers. Specifically, infrared markers were placed on each participant's (from the bottom to the top) lateral malleolus, lateral femoral condyle, greater trochanter, and at the mid-axillary line, just above the hip. A similar setup has been previously utilized in our research laboratory.[25,27] This setup allowed for the changes in the knee joint angle (lateral malleolus and greater trochanter rotating about lateral femoral condyle) and the hip joint angle (mid-axillary and lateral femoral condyle rotating about greater trochanter) to be recorded on the Codamotion software in real time. The markers were attached after the MVC tests. Codamotion data were collected at a sample rate of 1000 samples per second and were synchronized with the EMG data using a hardware trigger.

2.3 | Testing protocol

The data collection consisted of 10 repetitions of each of the three exercises of the L-protocol.[31,32] Participants performed the three exercises in a randomized order. The 10 repetitions were divided into two sets of five consecutive repetitions performed at a self-selected comfortable speed. Before the first set of repetitions of each exercise, participants were coached on how to perform the exercises correctly and their performance was monitored during a number of familiarization trials. In the EX exercise, participants were asked to lie on their back with their right leg flexed at 90° and held in place at the thigh using their own arms. Slow knee extensions were then performed to the point of maximal possible extension (that for our subjects was full knee extension, Figure 1). In the DV exercise, participants performed a simulated dive. They were asked to stand on the right foot, with the left knee bent at approximately 90° and the left foot raised, and to flex their hip while stretching their arms forward. Subjects were asked to attempt to achieve maximal hip extension of the raised leg while keeping the pelvis level. The right knee was kept slightly flexed at an angle of approximately 10° (Figure 1). In the GL exercise, participants were asked to stand on a slick wooden platform while holding a support bar on their left. They began the exercise standing in an upright position with their legs slightly split. Participants held on the support bar as their left leg glided backward. They were instructed to keep all of their weight onto the heel of their right leg. The motion was stopped at the point where the participant was not able to continue the split. The movement back up to starting position was performed using assistance of both arms.

2.4 | Data processing

All processing, data analysis, and statistical analysis were performed using custom code developed in MATLAB. The EMG signals were first off-line band-pass-filtered (Butterworth filter, third-order, zero-lag, cutoff frequencies 30 Hz and 450 Hz) to remove movement artifacts and unwanted components. The envelope of each EMG signal was then extracted by applying a low-pass filter (Butterworth filter, third-order, zero-lag, cutoff frequency 5 Hz) to the full-wave-rectified EMG signal. Each EMG channel was normalized to the median maximum EMG envelope obtained

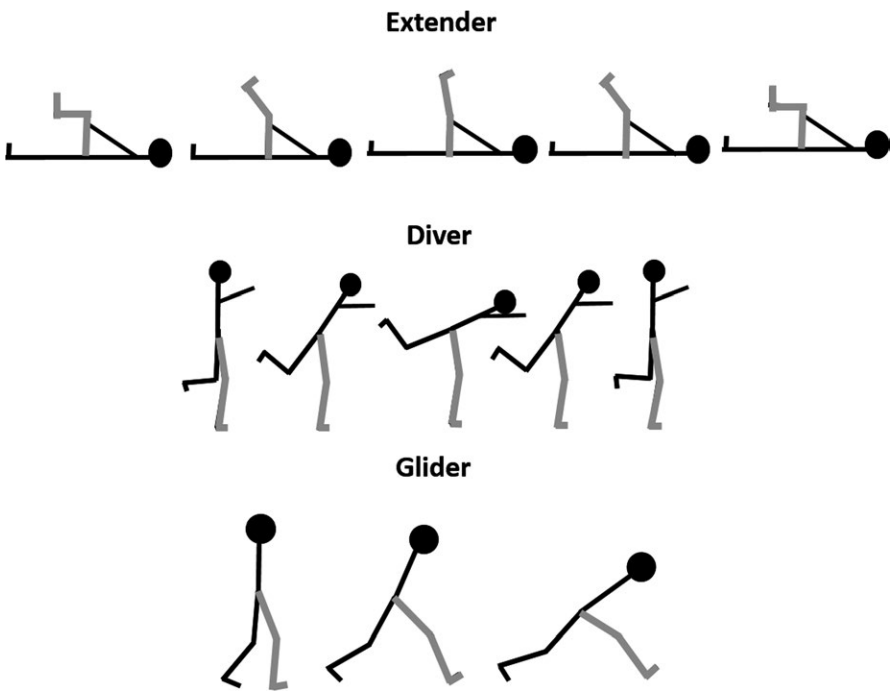

FIGURE 1 Graphical description of the exercises of Askling L-Protocol. In the extender (EX) exercise, the participant is required to fully extend the knee while keeping the hip flexed from a supine position. In the diver (DV) exercise, the participant is required to simulate a dive while standing on a leg. In the glider (GL) exercise, the participant is required to perform an assisted split while holding a handrail

from the three repetitions of the relative MVC exercise. Specifically, the RF was normalized using the knee extension MVC test, ST and BF using the knee flexion, and the GM using the hip extension test. The kinematic data were smoothed using a third-order Savitzky-Golay filter. The data (kinematic and EMGs) relative to each repetition of each exercise were then segmented and length normalized over 1000 samples. The EX exercise was segmented taking as a starting point the instant 200 ms before the knee flexion angle first fell below 65° and as a final point the instant 200 ms after the knee crossed again the 65° mark. The DV exercise was segmented taking as a starting point the instant 200 ms before the hip angle first reached 15° of flexion and as a final point the instant 200 ms after the hip crossed again the 15° mark. The GL exercise was segmented taking as a starting point the instant 200 ms before the hip angle first reached 25° of flexion and as a final point the instant 200 ms after the maximal hip angle recorded on the trial. For all three exercises, the eccentric phase of the hamstring muscles was identified. In the GL and DV exercises, the hamstring eccentric phase was, for each segmented repetition, defined between the beginning of the repetition and the instant of peak hip flexion. In the EX exercise, the hamstring eccentric phase was identified between the beginning of the exercise and the peak knee extension.

2.5 | Electromyographic and muscle synergy analysis

We compared the activity of the four muscles across the three exercises. To do so, we calculated the root-mean-squared (RMS) value of each EMG over each repetition. We performed a statistical analysis to assess for significant differences in the overall activation of the different muscles across the different exercises. In this analysis, we used a one-way ANOVA applied to the average (across repetitions) RMS value of each muscle across the different exercises. A post-hoc analysis based on Tukey's test was used to assess for specific differences between exercises. This analysis has been performed for the RMS calculated over the whole repetition of each task and also (for the BF and ST muscles) exclusively during the eccentric phase of the exercise. We used muscle synergy analysis (MSA)[35] to extract the co-activation patterns across the four muscles. MSA, in its simpler form, represents the activity of a set of muscles M as the activation of a smaller set of synergies (or modules) representing the co-activation patterns of the muscles (W) that are recruited together according to the activation patterns (H) that represent the single neural commands driving each synergy. The activity of the single muscles can be then reconstructed from the synergies and their activation patterns following the formula: $M \approx W \cdot H$. Each muscle can participate in more than one module W, and similarly, a module W can be constituted by just one or even by all the muscles under analysis. The modules W are considered the building blocks of the different movements and can be shared between different tasks.[36] The activation patterns are task specific and indicate the way in which each module is activated during a particular movement/exercise.

In the following, we will refer to the motor modules extracted from the MSA as co-activation modules (Co-AMs). In this analysis, we fixed the number of Co-AMs to be extracted using the nonnegative matrix factorization (NNMF) algorithm for all subjects[37] to 2, to reflect both the tonic and phasic components of muscular activation expected in the exercises. For this analysis, the amplitude-normalized EMG envelopes were fed to the NNMF in their entirety and before the segmentation and time normalization. The goodness of reconstruction was assessed by calculating the R^2 between the original envelopes and those obtained by multiplying the synergies modules and activation patterns. Statistical analysis based on one-way ANOVA was used to assess for statistically significant differences in R^2 of reconstruction across the three exercises.

3 | RESULTS

3.1 | Extender

In the EX exercise (Figure 2), the hip is kept fixed at around 60 degrees of flexion while the knee is extended from 85.45 ± 2.23 to 18.23 ± 2.17 degrees, with a peak knee velocity of 77.78 ± 6.76 degrees/s during extension and -101.90 ± 10.48 degrees/s during flexion. In this exercise, the RF drives the movement and reaches a peak activity of $23.53\% \pm 4.87\%$ MVC across all subjects (Figure 3) during the eccentric phase of the exercise relative to the hamstring muscles (concentric for the RF). The activity of the ST and GM muscles is negligible during this exercise, while the BF presents a constant tonic activity at $11.69\% \pm 2.73\%$ of MVC.

3.2 | Diver

The DV exercise (Figure 2) is characterized by a smooth flexion of the hip on the standing leg. The flexion reaches a consistent peak across subjects around 64.06 ± 2.00 degrees and a maximal hip angular velocity of 38.39 ± 2.81 degrees/s during the flexion phase of the exercise and -49.22 ± 3.81 degrees/s during extension. The knee remains slightly flexed around about 20 degrees during the whole exercise. During the DV exercise, the GM muscle reaches a peak (equal to $83.79\% \pm 12.81\%$ MVC) around the point of peak hip flexion (circa 70% of the length of the exercise, Figure 3). The hamstring muscles activate together with a profile only barely resembling the hip angle, with a peak activation below 50% MVC for both BF and ST around 90% of the exercise. The

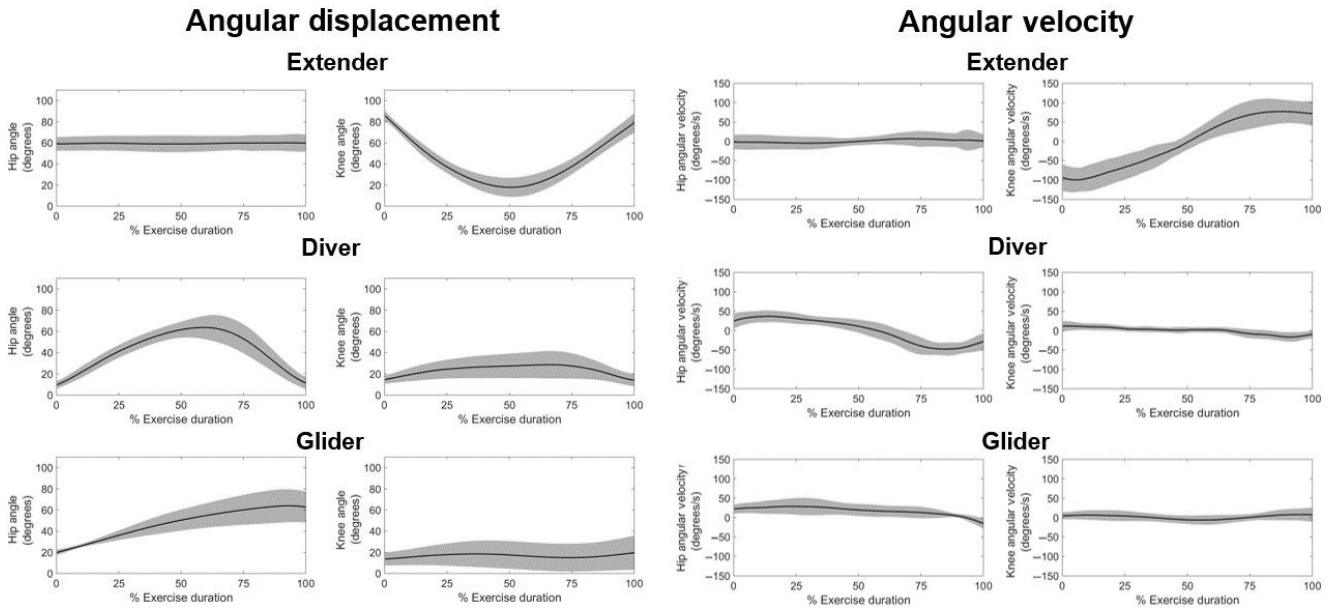

FIGURE 2 Kinematic analysis. Left panels show the angular displacements (hip left, knee right) for the three exercises of the L-Protocol (from the top to the bottom, extender, diver, and glider), while the right panels show the angular velocities. All plots present the average (bold black line) and standard deviation (gray shade) across subjects

peak activations of the BF and ST muscles occur at the beginning of the concentric phase of the exercise. During the eccentric phase of the exercise, the BF and ST reach maximal values of about 30% while the GM has peak activation of 60%, in all cases reached at the end of the eccentric phase. The activity of the RF muscle is negligible in this exercise.

3.3 | Glider

During the GL exercise (Figure 2), the hip flexion increases linearly from 19.94 ± 0.41 to 63.98 ± 3.55 degrees, with a maximal velocity of 29.96 ± 3.10 degrees/s. The hip angle increases up to 95% of the exercise, where a slight extensive movement can be observed. The knee is kept slightly flexed between 12.50 ± 2.04 and 20.52 ± 3.26 degrees during the whole exercise, with a slight increase in flexion toward the end of the exercise. The activity of the muscles of the hamstring increases over the course of the exercise, and the BF and ST muscles appear perfectly synchronized (Figure 3). The BF and ST muscles reach similar maximal contraction levels during the exercises (for both muscles between 55% and 60% MVC). Also, in this case, maximal activations are obtained at the beginning of the concentric phase of the exercise. At the end of the eccentric phase, the BF and ST muscles present maxima of about 35% MVC. The GM muscles share a profile of activation similar, but lower, to the one presented by the BF and ST muscles with a maximal value of about 28% MVC (about 18% during the hamstring eccentric phase), while the RF muscle maintains a tonic activation around 10% of the MVC (10.43% ± 0.86% MVC).

3.4 | Muscular activation across exercises

The RMS of each muscle was compared across the different exercises to assess for statistically significant differences (Figure 4A). The activity of the BF and ST muscles was consistent between the two muscles and statistically different across the exercises ($P < 0.01$, one-way ANOVA). Both muscles presented statistically significant higher values of RMS in the GL and DV exercises with respect to the EX exercise based on the post-hoc tests ($P < 0.01$, Tukey's test). No differences in activation were observed in the two hamstring muscles between the DV and GL exercises, indicating that the muscles maintain a similar level of activation. A similar, but opposite, pattern, was observed for the RF muscle, that is, a higher level of activity during the EX exercise that was statistically different than those observed in the DV and GL exercises ($P < 0.01$ in both the ANOVA and post-hoc). Once again, no differences were observed between the DV and GL exercises. The GM muscle presents significantly different levels of activation across the three exercises ($P < 0.01$ for one-way ANOVA and for all post-hoc comparisons). The GM was predominantly active during the DV exercise (about 50% of MVC across the exercise) while only limitedly active in the EX and GL exercises. The analysis on muscular activation during the eccentric phase of the exercise (BF and ST muscles, Figure 4B) yielded similar results as those performed on the whole exercises. Once again, we observed statistically significant higher activation of both muscles during the GL and DV exercises with respect to the EX exercise. The activity of the ST was not statistically different between

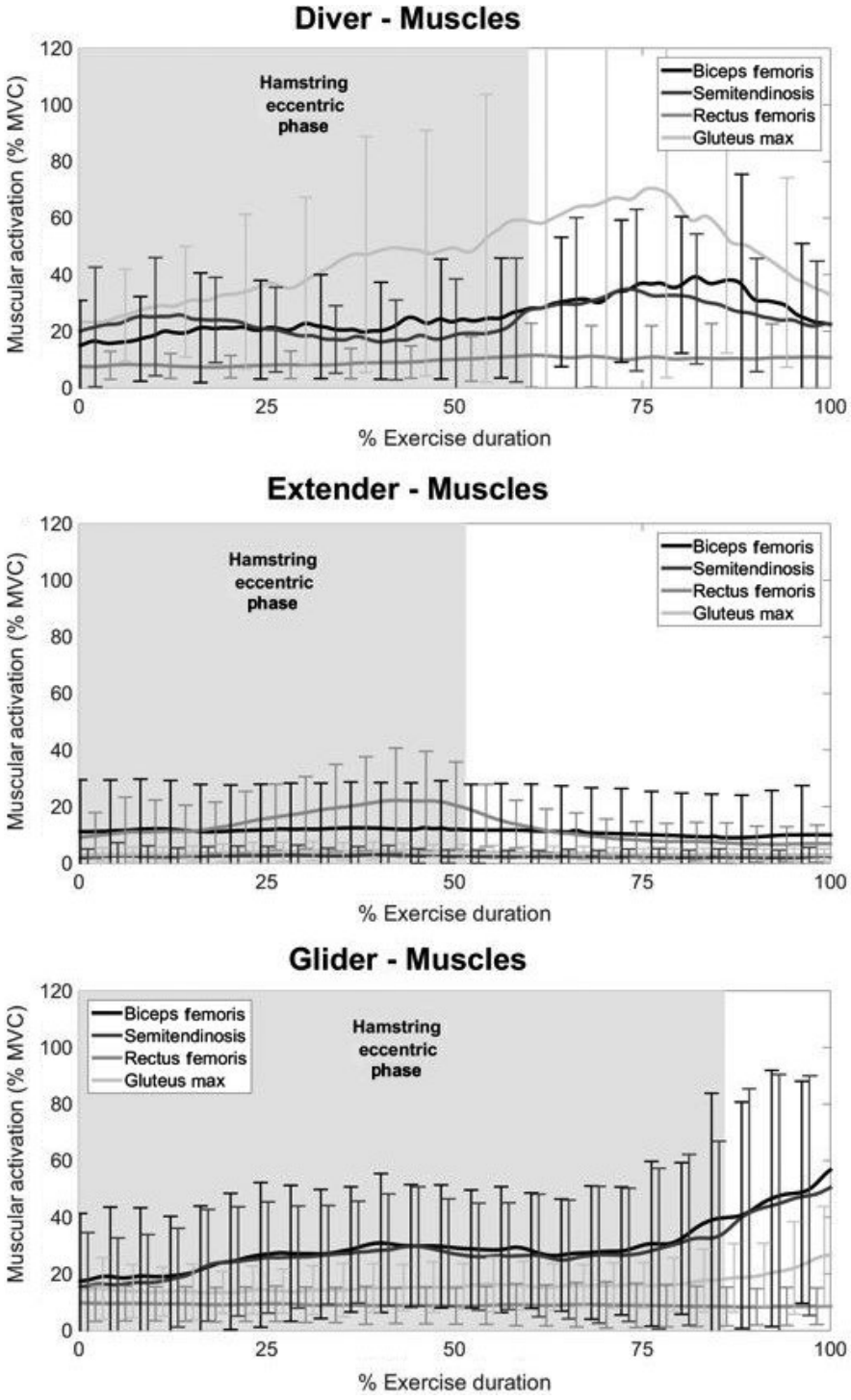

FIGURE 3 Muscular activation profiles. Activation profiles of the four muscles (BF, ST, RF, and GM in descending tones of color from black to light gray) across the four exercises (from the top to the bottom, diver, extender, and glider). All profiles are expressed as percentage of the MCV over the % exercise duration. Each plot presents the average and standard deviation of each muscle. In each plot, the gray shaded area represents the eccentric phase of the hamstring muscles during the exercise

the GL and DV exercises. On the other hand, we observed statistically significant higher activation of the BF during the GL exercise with respect to the DV.

3.5 | Synergies and co-activation analysis

The activity of the four muscles under analysis could be well explained through the activation of four different muscle Co-AMs (Figure 5). Co-AM1 and 2 were recruited during the DV and GL exercises specifically, the first one characterized by a prominent activity of the GM muscle, while the second one by the co-activation of the ST and BF muscles. In the DV exercise, Co-AM1, whose main contributor is the GM muscle, has a profile of activation that resembles the hip flexion profile, while in the GL exercise, it is activated steadily and at a lower level, possibly for stabilization purposes. Co-AM2, on the other hand, has a phasic activation in both the DV and GL exercises. In the former, the activation

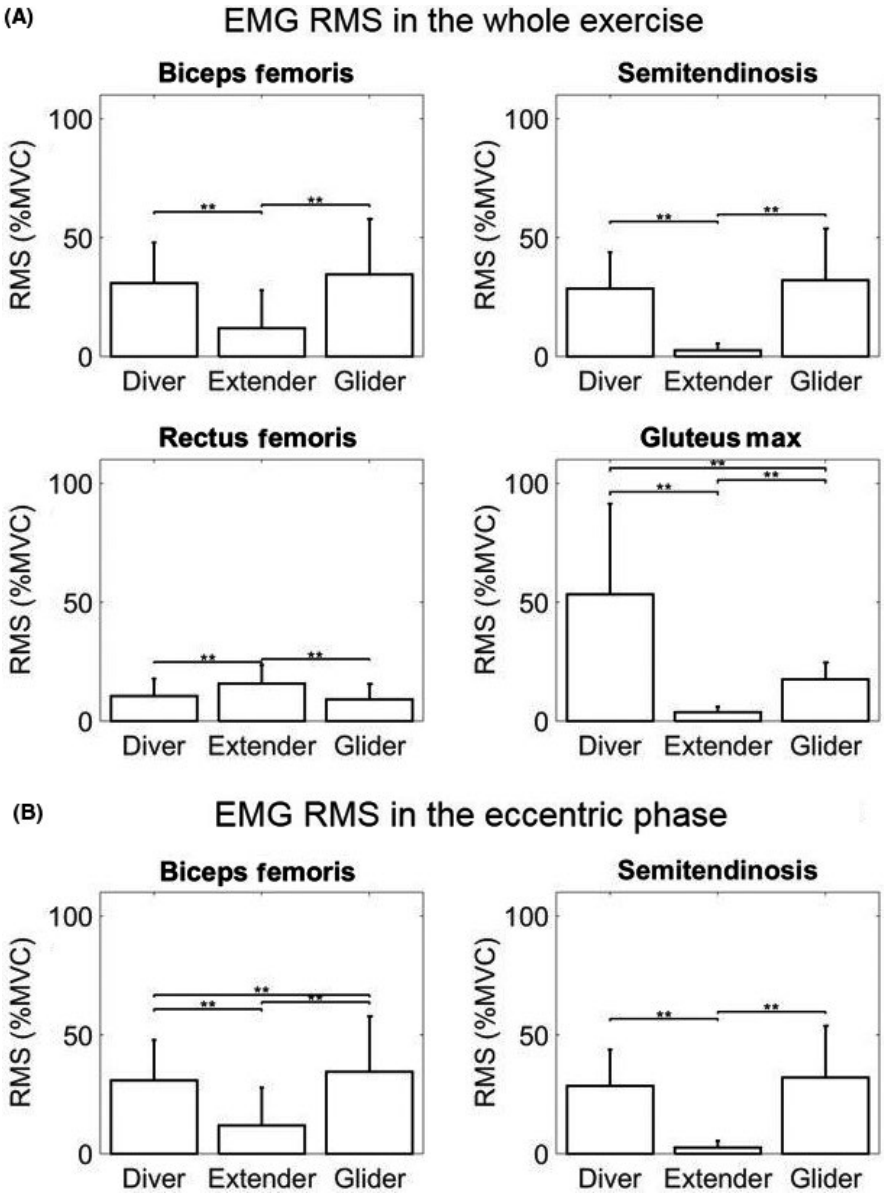

FIGURE 4 Bar graph of the RMS of each muscle across the different exercises. The RMS is presented as a percentage of the MVC. All statistical analyses were based on a one-way ANOVA for each comparison, with a post-hoc Tukey's test analysis. For all muscles, we observed p values from the one-way ANOVA test <0.01. The ** sign and the lines indicate which comparisons are significantly different following post-hoc analysis with $P < 0.01$. Panel A presents the results of the analysis performed calculating the RMS for all four muscles during the whole exercise. Panel B presents the analysis performed only on the BF and ST muscles during their eccentric phase

profile is characterized by a burst of activity toward 90% of the exercise, when the hip is starting to extend, while in the latter, the activity profile increases almost exponentially through the exercise. Co-AM3 and 4 were observed only during the EX exercise, the first one (Co-AM3) characterized by the phasic activity of the RF, while the second one (Co-AM4) characterized by the tonic activity of all the muscles together. For all three exercises, two Co-AMs could explain the variance of the dataset with an $R^2 > 0.95$. Statistically significant differences in the quality of reconstruction across the three exercises were not observed (one-way ANOVA, $P = 0.18$).

4 | DISCUSSION

In this study, we performed a kinematic and electromyographic characterization of the exercises of the L-protocol proposed by Askling et al[31,32] The L-protocol consists of a series of three exercises that differentially elicit eccentric contractions of the hamstring muscles. This protocol has been shown to significantly reduce the time to return to sport after hamstring muscle injuries when compared to a more "traditional" exercise-based rehabilitation protocol.[31,32] What emerged from our results is that in the L-protocol, the hamstring muscles are as follows: (a) contracted

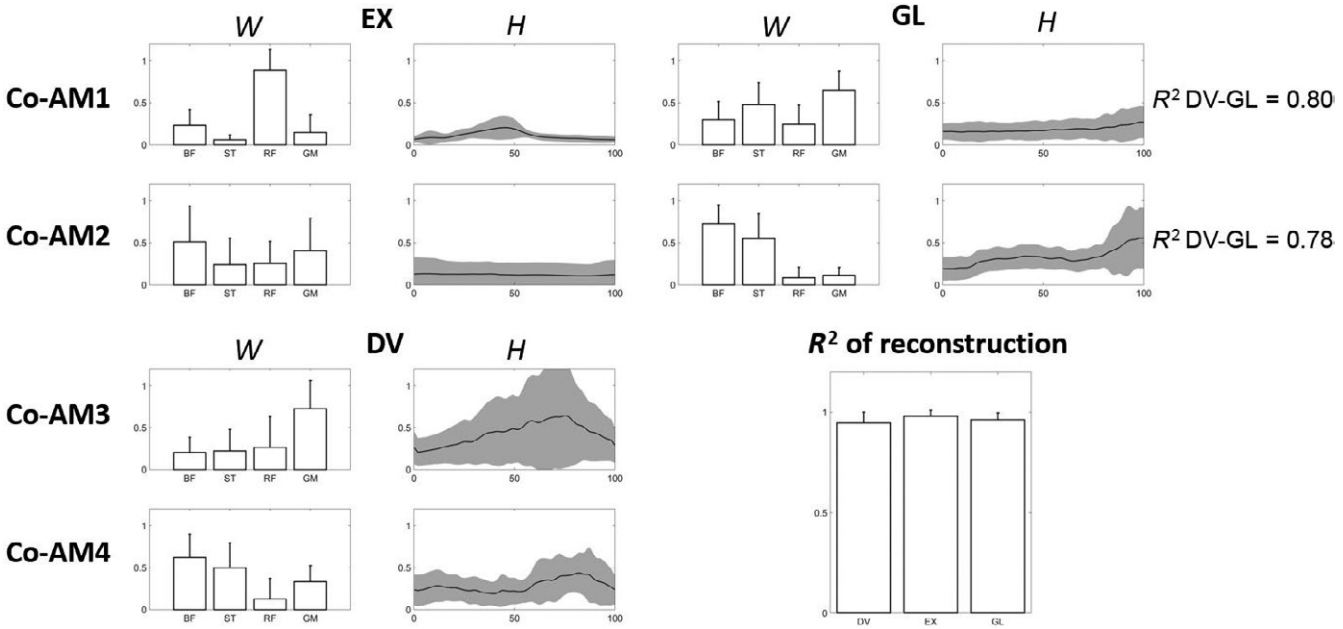

FIGURE 5 Co-activation analysis using muscle synergy analysis. Co-activation modules (Co-AMs) extracted from the different exercises are presented horizontally. The DV and GL presented the same Co-AMs (1 and 2, R^2 between the relative *Ws* of 0.80 for Co-AM1 and 0.78 for Co-AM2). The EX exercise presents two specific Co-AMs (3 and 4). The bottom right panel presents the R^2 of reconstruction using two modules for all exercises. Statistical analysis (one-way ANOVA) to assess for significant differences in reconstruction across the exercises did not yield significant results ($P = 0.18$)

eccentrically at a similar work point as they are during the swing phase in running and sprinting[15]; (b) passively stretched (EX) or actively eccentrically contracted to stabilize the body (DV) and resist hip flexion (GL); and (c) recruited in co-activation modules that can have different functional roles, either driving or stabilizing the movement, across the different exercises.

4.1 | Neuromechanical characteristics of the L-protocol

The key characteristic of the L-protocol is that all exercises include either an active lengthening (DV and GL) or a passive stretching (EX) of the hamstring muscles, based on the rationale that hamstring exercises performed at longer-than-standard muscle-tendon lengths could be beneficial when recovering from or when endeavoring to prevent hamstring injuries.[15]

These exercises were selected by Askling et al "… based on practical experience and applied anatomy, not on biomechanical analysis …"[31]; thus, the reasons behind their success in reducing time to return to sport have been, so far, evaluated only qualitatively. Askling et al[31] state that the three exercises load the hamstring muscles while accounting for their biarticular nature. Our results show that this assumption is partially true. In fact, in both the DV and GL exercises, while the range of motion (ROM) at the hip is substantial (>40 degrees for both exercises), the ROM at the knee is limited (<15 degrees). The opposite is true for the EX exercises, where the knee is far more engaged (ROM >50 degrees) while hip movements are negligible (<2 degrees). Thus, the hip and knee joints, although partially comoving in the DV and GL, are mainly engaged separately in the three exercises.

Our results show that only two of the three exercises lead to substantial (but submaximal) activation of the hamstring muscles. In the DV and GL exercises, the BF and ST muscles have an average activation level around 25% MVC and peak activations around 40% (DV) and 60% (GL) MVC during the whole exercise. During the eccentric phase of the exercises, both muscles present average activation around 20% MVC and peak activation around 25% (DV) and 40% (GL).

In the DV exercise, the activity of the GM muscle is the most prominent (average activation 47.50 ± 6.99, maximal activation 83.79 ± 12.81 of MVC) of the three posterior chain muscles analyzed, possibly reflecting the postural activation necessary to maintain balance and counteract gravity as the participant leans forward. In this exercise, the hamstring muscles present a roughly phasic profile of activation that poorly resembles the hip angle. This activation possibly reflects the muscle resistance to the passive stretch induced by the movement. In the GL exercise, the hamstring muscles actively control the execution of the movement, by regulating the speed of the split by resisting lengthening. This is reflected by the exponential increase in BF and ST activations through the exercise that continues even after the hip has stopped flexing (at around 95% of

the task). It is interesting to notice that in both the DV and GL exercises, the peak BF and ST activations happen while the hip is starting to extend, similar to what has been shown during sprinting and running.[15] The maximal muscular activation levels observed in the DV and GL exercises are, however, likely smaller than those expected during sprinting and running, where values above the MVC calculated during isometric contractions (as in our case) have been often reported.[38,39]

The aim of the EX exercise in the L-protocol is that of increasing hamstring flexibility through passive, unresisted stretching. The hamstring muscles in this exercise present a constant activation profile at low MVC levels (~10% for BF and ~3% for ST) that reflect a role in the stabilization of the hip while the RF performs the knee extension movement. The fact that the hamstrings do not present an increased activation profile during the knee flexion part of the exercise suggests that the participants preferably achieve a smooth knee flexion by resisting gravity through the activation of the knee extensors.

Another commonly utilized eccentric exercise for the hamstring muscles is the Nordic hamstring exercise (NHE).[17] The mechanical effect of the NHE strength training is that of shifting the muscle's torque-angle curve, allowing for the hamstring to generate peak force at increased muscle lengths.[17] In a recent study, we demonstrated that this is correlated with an increased ability to control the downward fall through the activation of the hamstrings.[27] Our results suggest that the GL exercise is conceptually similar to the NHE, whereby the eccentric contraction of the lengthening hamstring is used to resist and indirectly control a gravity-actuated movement. However, it needs to be pointed out that the hamstring lengthening is obtained differently in the NHE with respect to the GL. During the NHE, in fact, the hips are stationary while the knee is extending, while the opposite happens during the GL. Moreover, the NHE elicits higher levels of hamstring activation (close to 100% MVC[40]) with respect to the GL exercise. This is likely due to the fact that during the NHE, the hamstrings are activated to resist the fall of the upper body, while in the GL, they resist the sliding of the leg. In the NHE, the BF and ST have been shown to present different activities.[41] The exercises of the L-protocol, on the other hand, seem to elicit consistent activity between the BF and ST muscles, with only slightly higher levels of activation of the BF during the glider exercise.

4.2 | Muscle synergy analysis

In this work, we used muscle synergy analysis (MSA) as an index of muscular co-activation across the different exercises. As MSA is commonly used to describe the neurophysiological strategies of muscular co-activation, it is preferably performed on large datasets recorded from several muscles ($n > 7$) during different movements.[42] The analysis herein presented does not claim to have the "neurophysiological" validity commonly associated with MSA and should not be interpreted accordingly, but it is instead utilized to derive an easily understandable index of functional co-activation of the four muscles we analyzed during the exercises.

The premise of our MSA-based co-activation analysis is that in the exercises we analyzed, a subset of the muscles would act as "drivers" for the movement, while another subset would act as "stabilizers," with potential overlap between these two groups, in terms of both muscles and functionality. This rationale is reflected in the consistent selection of two synergies as input for the factorization algorithm. Although we performed MSA separately for the three different tasks (due to the limited number of muscles available), we were able to observe that the shapes of the Co-AMs we found for the DV and GL are remarkably similar (median $R^2 = 0.80$ for Co-AM1 and =0.78 for Co-AM2). On the other hand (and as expected), the shapes of the Co-AMs extracted from the EX exercise are unique to that exercise. This result suggests that the three exercises employ four separate motor co-activation patterns (two shared between the GL and DV exercises and two specific to the EX exercise). It has to be noted that by pooling the data of the three exercises together for the MSA, we would have been able to find only a maximum of three unique co-activation patterns.

The MSA highlighted an interesting difference between the GL and DV exercises. In fact, although the two exercises are characterized by the recruitment of the same two Co-AMs (1 and 2), those modules appear to have different functions between the two exercises. In the DV exercise, in fact, the progression in hip flexion is paired with the similarly shaped activation of Co-AM1, indicating that this module acts as "driver" (or "active resistor") for the exercise, while the hamstring-focused Co-AM2 appears to act as "stabilizer" across the exercise. For the GL, this situation is reversed. In this exercise, in fact, the activation of the hamstring module Co-AM2 more closely resembles the progression in hip angle observed in the kinematic analysis, while Co-AM1 presents a fairly steady level of activation that suggests a stabilizing task for this module. Thus, although in the GL and DV exercises, we observe similar levels of activation (and co-activation) of the hamstring muscles (Figures 4 and 5), the functional role of these muscles across the exercises is different.

5 | PERSPECTIVE

In this work, we characterized the neuromechanical profile of the exercises of the L-protocol. What emerged from our results is that the L-protocol, through its three different exercises, elicits different beneficial effects in

the hamstrings. In the exercises, the hamstrings are either passively stretched or eccentrically contracted and trained to contract at a work point similar to that achieved during running[15] and during comparable exercises (eg, NHE[40]), but at lower overall activation levels with respect to both aforementioned cases. We observed a conceptual similarity between the GL exercise and the NHE[27] that may suggest a similar effect on shifting the muscle's torque-angle curve,[17] although our results cannot confirm this point. What appears clear from this work is that the advantage of the L-protocol with respect to a more "traditional" exercise-based rehabilitation protocol is that of building strength in the muscles while they work, in different ways, similarly (although with smaller activation levels) as they do in the most common injury situations. The combination of the similarity in eccentric activation work point and the lower activation profiles obtained during the exercises of the L-protocol is possibly correlated with the lower recurrence of injury associated with these exercises.

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